

Joonsuk Bae

Ph.D. Candidate, Sungkyunkwan University • ALICE & ePIC Collaborations

✉ jbae@cern.ch 🌐 joonsukbae.com 📄 0009-0008-4806-8019 📧 joonsukbae INSPIRE-HEP

Lead analyzer of the first ALICE Run 3 charged-particle jet measurement. Pb/scintillating-fiber calorimetry R&D for the ePIC Barrel Imaging Calorimeter.

Education

Ph.D. Candidate, Physics, Sungkyunkwan University 2022.03 – expected 2026.08

Advisor: Beomkyu Kim. Thesis: *R-dependent charged-particle jet production in pp at $\sqrt{s} = 13.6$ TeV with ALICE.*

B.S., Physics, Sungkyunkwan University 2021.02

Appointments and Collaboration Membership

ALICE Collaboration, CERN 2022.10 – present

ePIC Collaboration, BNL 2024.02 – present

LAMPS Collaboration, RAON 2022.03 – 2024.04

Research Highlights

- **Lead analyzer**, first ALICE Run 3 charged-particle jet cross-section measurement at $\sqrt{s} = 13.6$ TeV (in preparation, target 2026); full pipeline from O²Physics task development to systematics and unfolding.
- **PWG-JE Run 3 tracking-systematics task force** (convened 2026.02 by H. Caines and L. Havener, Yale; biweekly): pp 13.6 TeV lead analyzer; derived the total charged-particle tracking-efficiency systematic ($\sim 1.5\%$ across the dominant p_T range); ITS-TPC matching code adopted by J. Lee for PbPb.
- **ALICE O²Physics framework** (AliceO2Group/O2Physics, 35 PRs / 19 merged): central patch in Common/ (PR #15174, TrackTuner) and sole author of `jetCrossSectionEfficiency.cxx` (PR #14057) used by all Run 3 charged-jet analyses for normalization.
- **ePIC Barrel Imaging Calorimeter**: lead offline analyst across CERN PS T10 (2025.07) and KEK AF-AR (2025.03) test beams of the Pb/scintillating-fiber prototype; JANA2 reconstruction studies for sampling-fraction and energy response.
- **Mentoring**: PWG-JE Run 3 O-O charged-jet measurement led by junior co-analyzer W. Ham in the SKKU group – mentored on jet reconstruction methodology and systematic uncertainties; co-author on the corresponding Analysis Note.

Selected Publications

Full list: inspirehep.net/authors/2117232 · ALICE co-author since 2022.10 (**109 collaboration papers**).

- **FLAGSHIP** ALICE Collaboration, *Inclusive and semi-inclusive charged-particle jet cross sections in pp collisions at $\sqrt{s} = 13.6$ TeV* (in preparation, target 2026). [*lead analyzer*]
- J. K. Adkins et al. (ECCE Collab.), “Design of the ECCE detector for the EIC,” Nucl. Instrum. Meth. A **1073** (2025) 170240 (> 280 cites). [*co-author*]
- S. Y. Jang, G. An, **J. Bae** et al., “Performance of dual-readout calorimeters for various absorbers,” Nucl. Instrum. Meth. A (2026). [*co-author*]
- G. Cho, S. Kim, **J. Bae** et al., “Beam tests of the copper-based dual-readout calorimeter for e^+e^- colliders,” J. Subatom. Part. Cosmol. **3** (2025) 100021. [*co-author*]
- Y. J. Kim et al. (LAMPS), “Large Acceptance Multi-Purpose Spectrometer (LAMPS) at the RAON,” J. Korean Phys. Soc. **87** (2025) 670–682. [*co-author*]
- Two submitted (NIM-A): *AstroPix silicon test-beam (NIMA-D-26-00572) and Pb/SciFi BIC beam-test (NIMA-D-26-00482).*

Invited and Contributed Talks

Summary since 2022: 3 invited workshops · 5 invited collaboration-meeting talks · 6 international conference contributions (one on behalf of ALICE at EPS-HEP 2025) · 7 domestic · PWG-JE internal: 50+.

Selected:

- **EPS-HEP 2025**, Marseille [*on behalf of ALICE; PWG-JE merge talk*] – 2025.07
- **HIM 2025**, APCTP [*invited workshop*] – *From precision to structure: jet measurements in ALICE Run 3* – 2025.05
- **Quark Matter 2025**, Frankfurt [*international, poster*] – 2025.04
- **ALICE Physics Week 2024** [*invited collab meeting*] – *Charged-particle jet production in Run 3* – 2024.12
- **Hard Probes 2024**, Nagasaki [*international, poster*] – 2024.09

Technical Skills

Programming: C++, Python, ROOT, Geant4, JANA2/EIC-recon, PYTHIA, FastJet, Bash, Git.

Analysis: jet reconstruction and unfolding (Bayesian, SVD), efficiency and systematic-uncertainty evaluation, test-beam calibration.

Detector hardware: PMT and SiPM readout characterization, optical-coupling QA, calorimeter module assembly.

Languages: English (fluent), Korean (native).

References

Available on request.